

Cassandra Vandevoorde

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RESEARCH INTERESTS

- Learning and memory through electrophysiology techniques (elaborate)
- Keywords: Electrophysiology, Hippocampal ripples, Learning and Memory, Mouse-model, high density probes, chronic recordings, acute recordings

EDUCATION

Agnes Scott College Decatur, GA *Bachelor of Science in Neuroscience, AI Minor* Expected May 2027

Relevant Coursework: Systems Neuroscience, Cellular Neuroscience, Neuroendocrinology, Integrative Biology, Cognitive Biology, Developmental Biology, Gen Chem, Organic Chemistry, Calculus I, Research Statistics, Research Design and Methods, Intro to Computer Programming, How to think like a Data Scientist

RESEARCH EXPERIENCE

Haider Lab / Georgia Institute of Technology Atlanta, GA

Student Researcher, Advisor: Bilal Haider

June-August 2026

- Assisted on successful craniotomies, perfusions, and brain extraction procedures on mice
- Quality checked and trained DeepLabCut model tracking pupil area, eye area, and motion energy
- Coded MatLab scripts used to create novel figures comparing changes in pupil area and motion energy between an ASD and WT mouse model

McNaughton Lab / UC Irvine Summer Institute in Neuroscience Irvine, CA

SF-funded Student Researcher, Advisor: Bruce McNaughton June-August 2025

- Performed successful craniotomies, perfusions, and brain extraction procedures on multiple mice
- Aided in electrophysiology recordings using head fixed mice on a novel treadmill set up with VR
- Coded several scripts used to create hundreds of novel figures and raster plots representing neural spiking data in the HPC, PPC, and S1 regions
- Discovered S1 narrow interneurons showed peak activity ~0.35 sec after HPC peak ripple times

Systems in Neuroscience Research Lab / Agnes Scott College Atlanta, GA

Research Course Student, Independent project

January-May 2025

- Earned CITI (Collaborative Institutional Training Initiative) Research Certification needed for humane handling, ethical protocol compliance, and research procedures involving murine models.
- Designed and executed experiment to test anxiety and depression-like behaviors in mice exposed to a 12 hour blue light and dark cycle when compared to a control group white light and dark cycle, gaining skills in rodent husbandry and light automation
- Found and presented significant results linking blue light exposure to a decrease in anxiety-like symptoms in female mice

Cellular Neuroscience Research Lab / Agnes Scott College Atlanta, GA

Research Course Student, Independent project

January-May 2025

- Earned CITI (Collaborative Institutional Training Initiative) Research Certification coursework needed for ethical care and handling of invertebrates
- Designed and completed experiment to test threat and fight behaviors in crayfish exposed to L-theanine when compared to a control group, gaining skills in crayfish care
- Found and presented significant results linking L-theanine exposure to decreased fight duration in crayfish

Developmental Biology Research Lab / Agnes Scott College Atlanta, GA

Research Course Student, Independent project

August-December 2025

- Designed and completed an experiment to compare effects of mineral sunscreen to reef safe sunscreen on sea urchin development, aiming to provide unbiased findings to inform consumer decisions
- Found and presented results showing urchins in water contaminated with mineral sunscreen did not reach second or third

division while urchins in water contaminated with reef-safe sunscreen successfully reached third division

SCALE Professional Development Program / Microsoft Atlanta, GA

Intern

March 2025

- Pitched and coded an AI power mentor matching system
- Worked with various teams at the Atlanta office to improve SCALE program, spearheaded addition of a Robotics Lab workshop

Eyewire Predictive Model-Based Labeling Virtual

Citizen Science Volunteer

August 2023

- Aided the Princeton Seung Lab in extracting brain structure from microscopic images by tracing neurons to train a deep convolutional network.
- Scanned and traced 24 samples using Eyewire interface, learning about the value citizen science projects and connectomics

Honors: Agnes Scott Dean's Scholarship (~\$24,000/year), Scottie Scholar Grant (\$1,000/year), Alumnae Scholarship (\$5,000/year), Dean's Honor List, NSF-funded scholar (~\$7,000), 2026 Goldwater Nominee

SKILLS

- *Tools:* Google Sheets, BioRender, Canva, Scientific Diagramming (Procreate), MatLab, Prism, Python ● *Lab Skills:* Light and confocal microscope, field data collection, live animal handling, titration, pipetting, rodent surgery, perfusion, brain extraction, cryostat, mounting and staining
- *Languages:* Spanish (Fluent), French (Fluent), English (Fluent)

PRESENTATIONS

Delvira Vandevoorde, C., Ning, W., Mcnaughton, B. "Cortical dynamics in PPC and S1 during awake rest and HPC ripples." *Oral& Poster Presentation, Summer Institute in Neuroscience symposium, August 2025.*

Delgado, E., Delvira Vandevoorde, C., Robinson, J., Dutton, D. "The Effects of Blue Light on Anxiety-like and Depression-like Symptoms in Mouse Models." *Poster Presentation, Scotties with Nerves, Agnes Scott College, April 2025. (Best in Neuroscience, Agnes Scott College)*

Delvira Vandevoorde, C., Boyd, N., Dutton, D. "The Effects of The Effects of L-Theanine on Threat and Flight Behaviors in Crayfish Models." *Poster Presentation, Scotties STEM Poster night, Agnes Scott College, December 2025.*

Carter, P., Gehner, L., Hughes, Delvira Vandevoorde, C., Larimore, J. "The Effects of Reef-Safe Sunscreen on Embryonic Development in *L. variegatus*" *Poster Presentation, Scotties STEM Poster night, Agnes Scott College, December 2025.*

LEADERSHIP AND COMMUNITY INVOLVEMENT

Math Club / Agnes Scott Atlanta, GA

President

May 2025 - Present

- Recruited and presented at college club fair, recruiting over 10% of the incoming class
- Ran weekly meetings reviewing Putnam problems and advanced math topics in order to make the putnam competition more accessible to students across disciplines.
- Spearheaded, planned, advertised and hosted STEM Survive and Thrive night, an event providing networking opportunities for all the stem organizations on campus as well as mentorship for undergraduates on CVs, Personal statements, REU applications, and alumni connections, as well as free headshots. Over 25% of attendees reached out again for mentorship and help applying to their first research experiences.

Treasurer

December 2024 - May 2025

- Created an approved budget proposal for 2025-26 academic year based on previous years spendings, accounting for planned events, weekly meetings, and guest speakers.

Strategies for Ecology Education Diversity and Sustainability (SEEDS) / Agnes Scott Atlanta, GA

Executive Board Member

September 2023 - May 2025

- Examined non-vascular plants
- Aided in the transportation of specimens from the Biology Research Station
- Planned, marketed and organized an educational ecology related event every semester open to the community. These include opportunities to propagate plants, learn about light and moisture requirements, and the benefits of growing native plants

PsychNeuro Club @ Agnes Scott Atlanta, GA

Neuroscience chair

August 2024 - Present

- Recruited and presented at college club fair
- Facilitated communication with Neuroscience Department
- Advocated for Neuroscience events and resources as the first person to hold the position

Treasurer

August 2026 - Present

- Attended IOC meetings and managed 2026-2027 budget

OTHER WORK EXPERIENCE

Registrar's Office / Agnes Scott

Paid Student Archivist January 2024 - Present ● Organized and preserved over 7 decades and over 20,000 files of student and alumni records ● Spearheaded the relocation and reorganization of Alumni records, implementing a new filing system that significantly improved efficiency in locating student files.

- Assisted in the successful launch of a new student data registration program, ensuring smooth transition and minimal disruption.

Axiom Healthcare Services/ Princeton, NJ

Clinical Trial Intern

June - August 2024

- Aided in the development of an educational session on Minimal Residual Disease (MRD), including reviewing and analyzing recent clinical trial data and FDA guidelines to ensure the accuracy and relevance of the content.
- Reviewed and organized information on 36 ongoing Stage 3 clinical trials for Amyotrophic Lateral Sclerosis (ALS) and Alzheimer's Disease to create an infographic for potential stakeholders.

REFERENCES

*in alphabetical order

- **Dr. Bruce McNaughton (Research Mentor)**
Professor, Center for Neurobiology of Learning and Memory
University of California, Irvine
Email: brucemcn@uci.edu
- **Dr. Amanpreet Kaur (Internship Mentor)**
Competitive Intelligence, Manager
Galapagos